



(12) **United States Patent**
Ding et al.

(10) **Patent No.: US 12,413,741 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **METHOD AND APPARATUS FOR VIDEO CODING PERFORMANCE ASSESSMENT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/621,713**

(22) Filed: **Mar. 29, 2024**

(65) **Prior Publication Data**

US 2025/0024042 A1 Jan. 16, 2025

Related U.S. Application Data

(60) Provisional application No. 63/527,060, filed on Jul. 16, 2023.

(51) **Int. Cl.**
H04N 19/154 (2014.01)
H04N 19/105 (2014.01)
H04N 19/149 (2014.01)

(52) **U.S. Cl.**
CPC **H04N 19/149** (2014.11); **H04N 19/105** (2014.11); **H04N 19/154** (2014.11)

(58) **Field of Classification Search**
CPC .. H04N 19/149; H04N 19/105; H04N 19/154; H04N 17/004

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2006/0233448 A1* 10/2006 Pace G06V 10/755 382/236
2015/0092861 A1* 4/2015 Sjoberg H04N 19/174 375/240.18
2018/0278934 A1* 9/2018 Andersson H04N 19/196
2018/0332281 A1* 11/2018 Yu H04N 19/176

OTHER PUBLICATIONS

Thorsten Laude et al., “a Comprehensive Video Codec Comparison” Cambridge University Press, doi:10.1017/ATSIP.2019.23 (Year: 2019).*

International Search Report and Written Opinion issued in International Patent Application No. PCT/US 24/30529 dated Aug. 21, 2024, 11 pages.

Chen et al. “An overview of coding tools in AV1: the first video codec from the alliance for open media.” APSIPA Transactions on Signal and Information Processing 9, e6, Jan. 9, 2020.

* cited by examiner

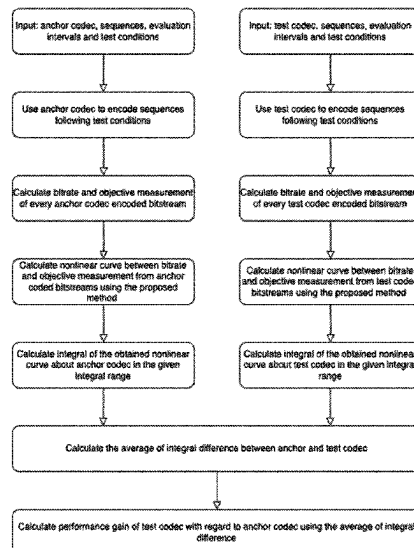
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(57) **ABSTRACT**

Methods, apparatus, and computer readable storage medium evaluating codec performance. One method includes obtaining m anchor data points each generated based on a respective anchor encoded video bitstream; obtaining n test data points each generated based on a respective encoded test video bitstream, n being an integer; fitting the m anchor data points with an anchor curve, the anchor curve being based on an anchor polynomial, wherein the anchor polynomial is monotonic in an x-axis range; fitting the n test data points with a test curve, the anchor curve being based on a test polynomial, wherein the test polynomial is monotonic in the x-axis range; and evaluating the test codec performance based on the anchor curve and the test curve, to obtain an evaluation result.

20 Claims, 11 Drawing Sheets



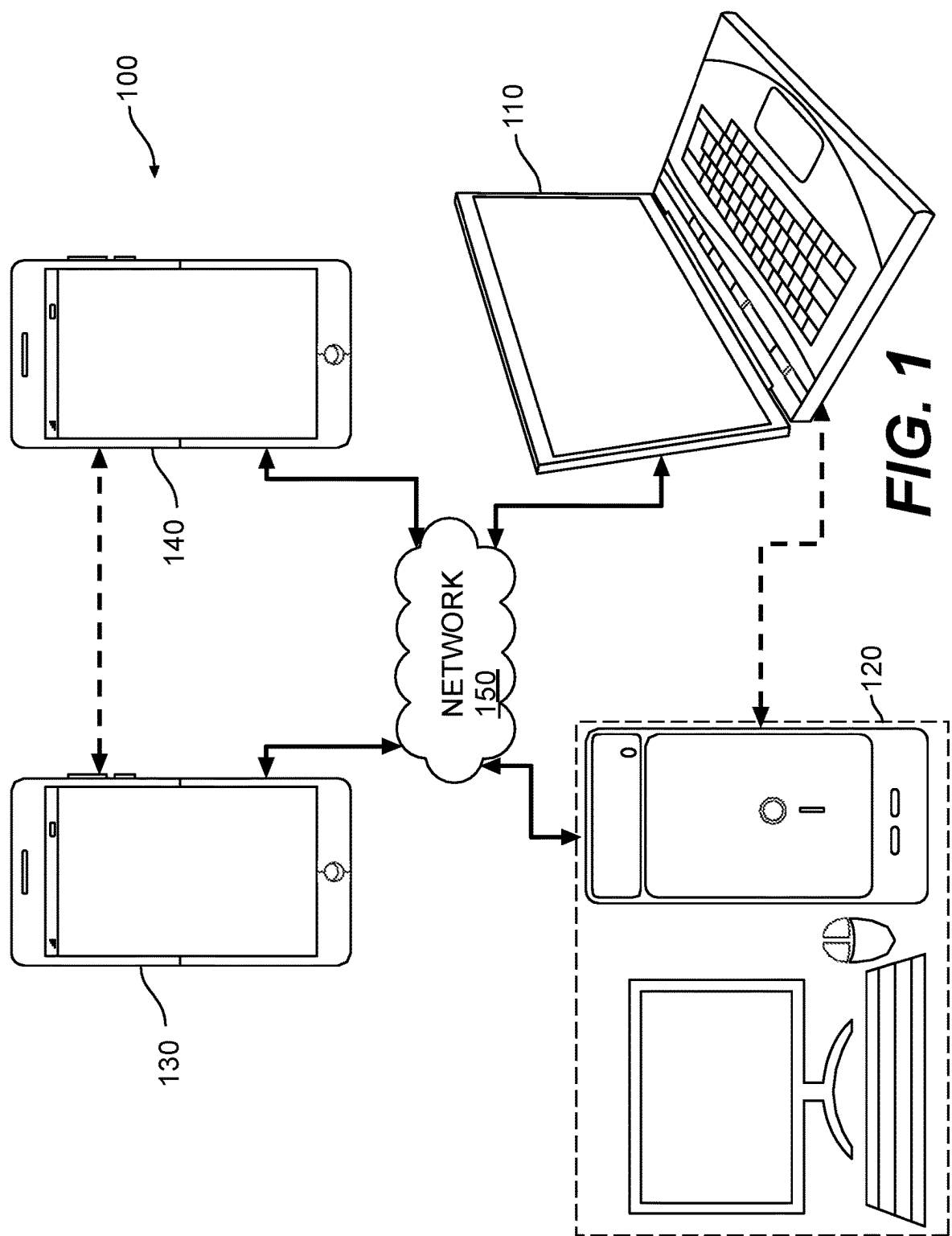


FIG. 1

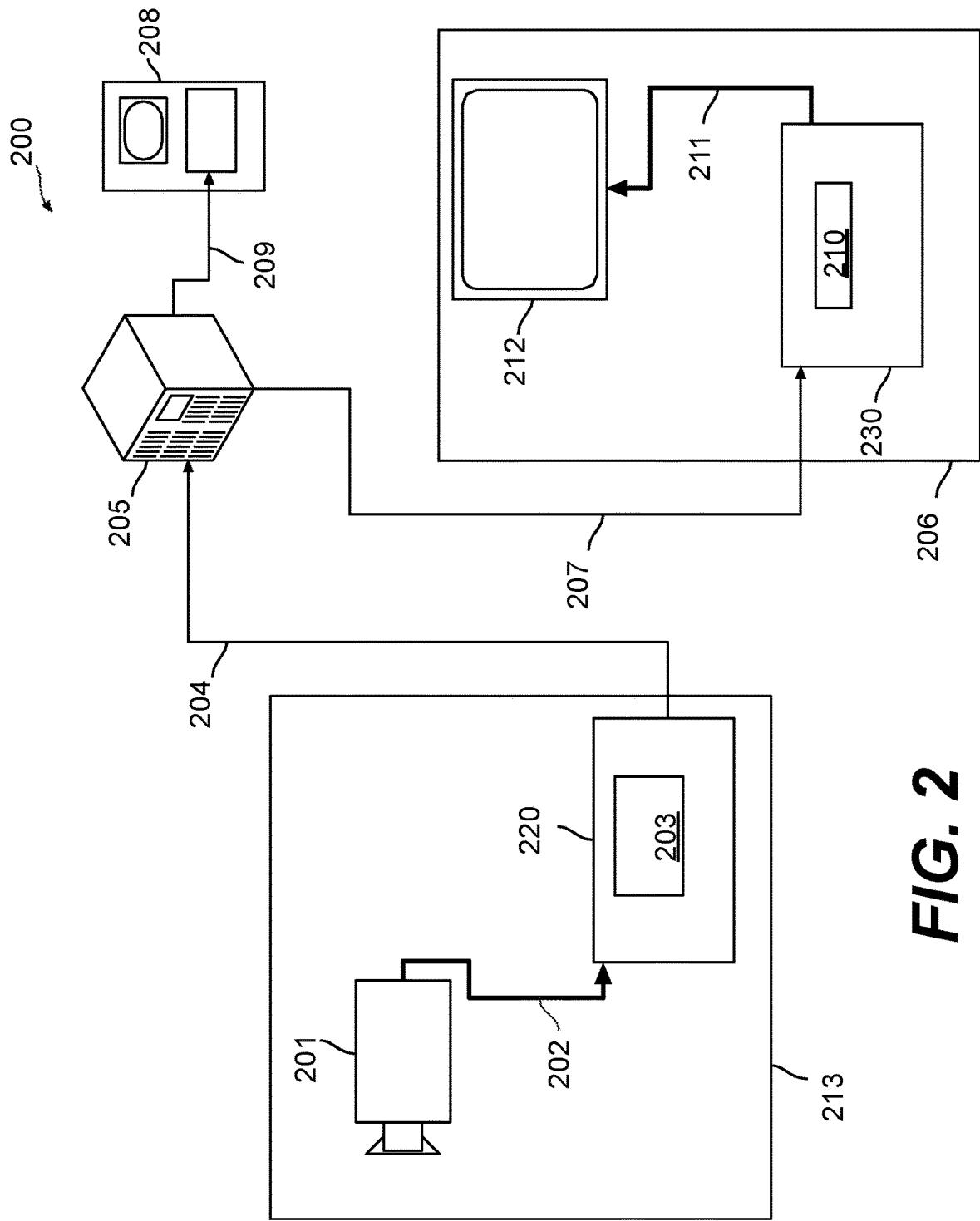


FIG. 2

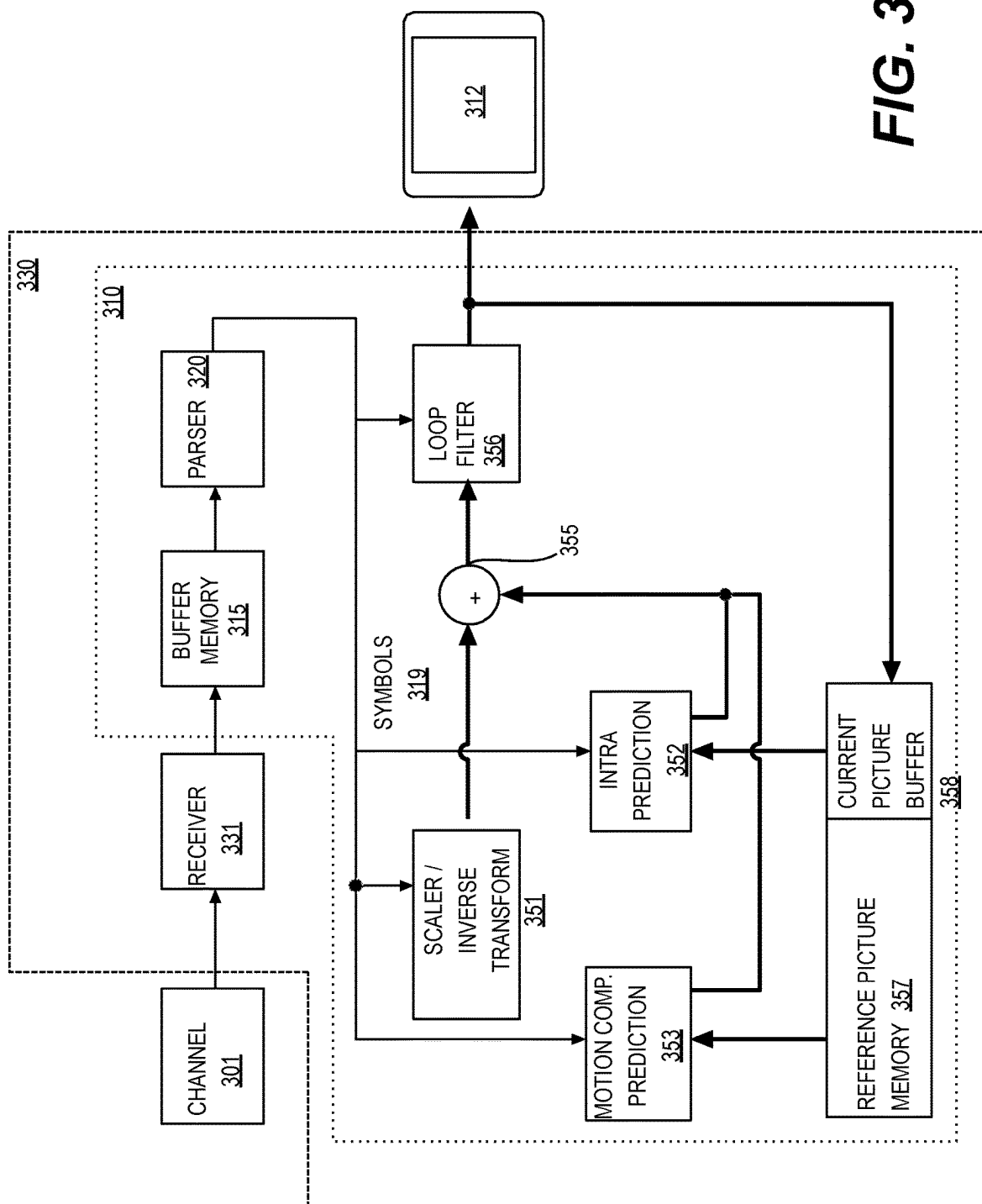


FIG. 3

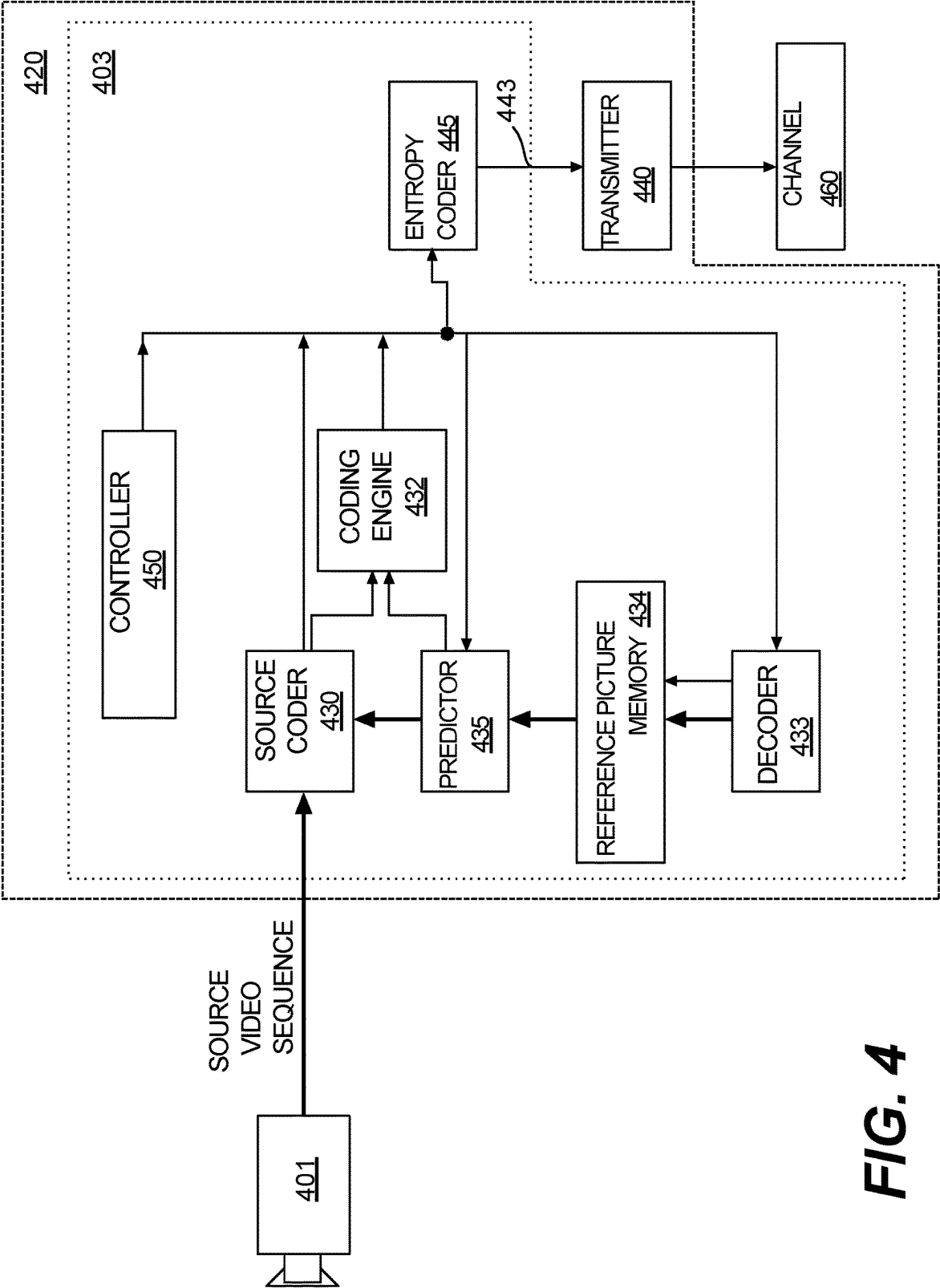


FIG. 4

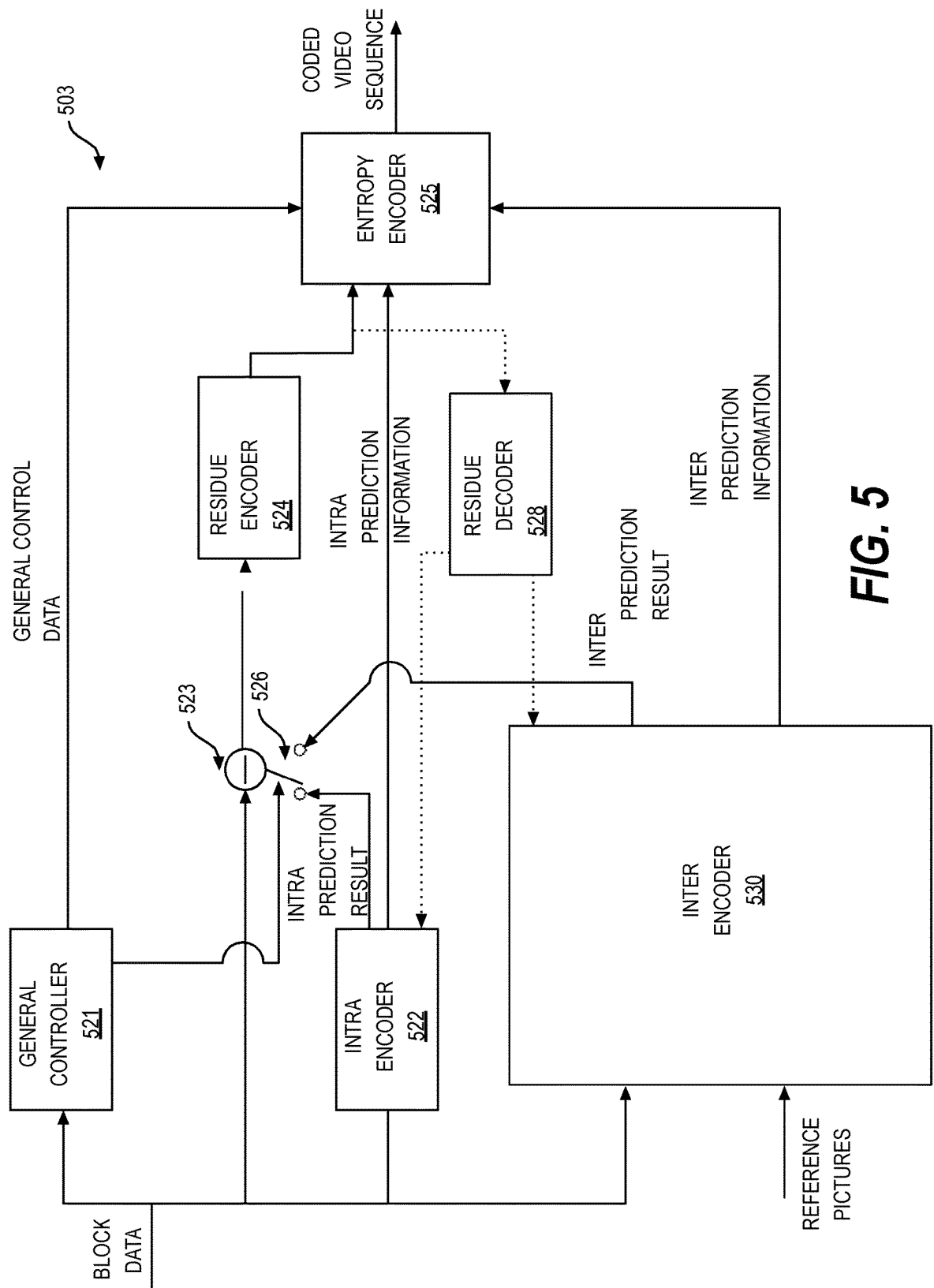


FIG. 5

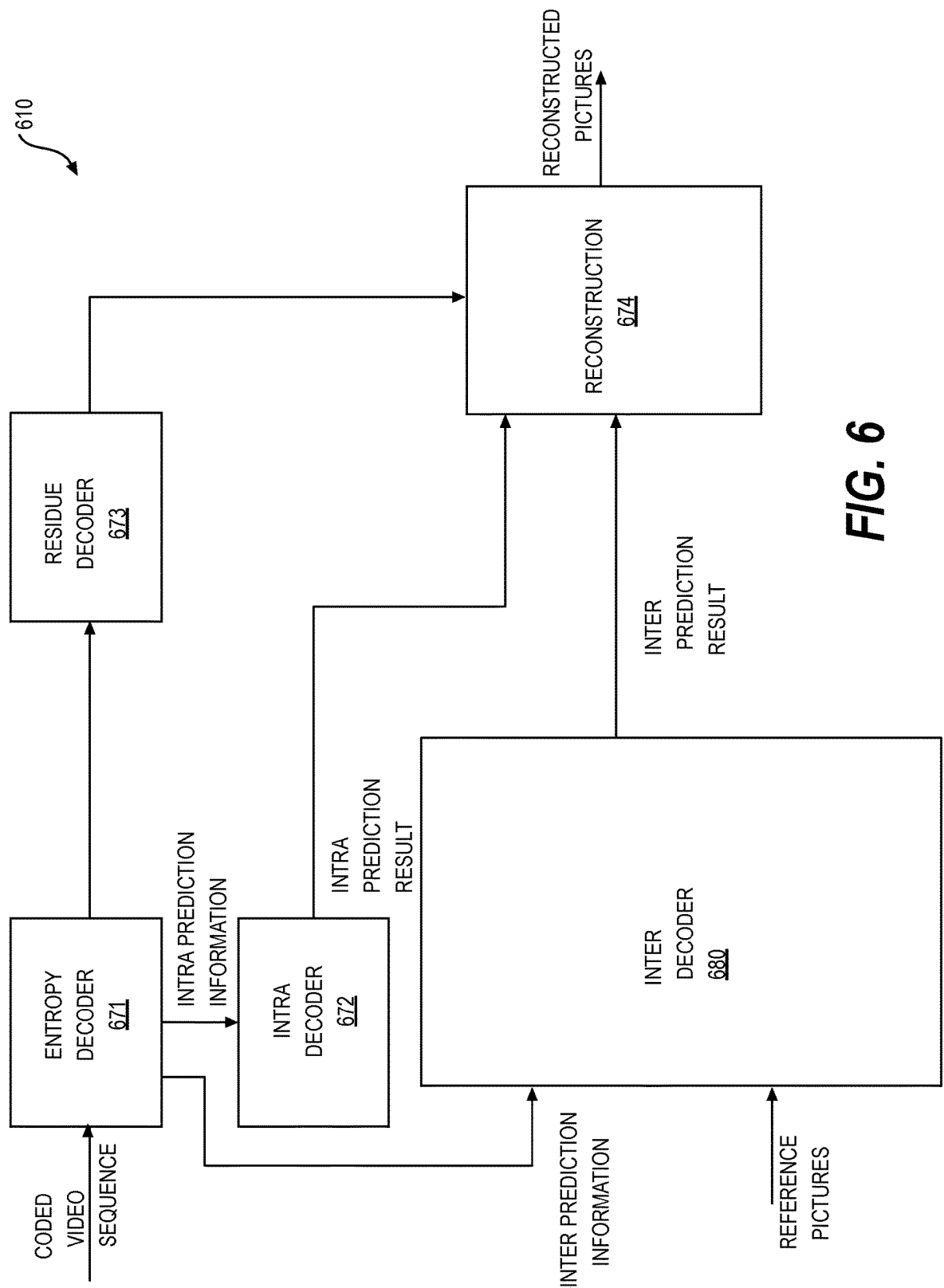


FIG. 6

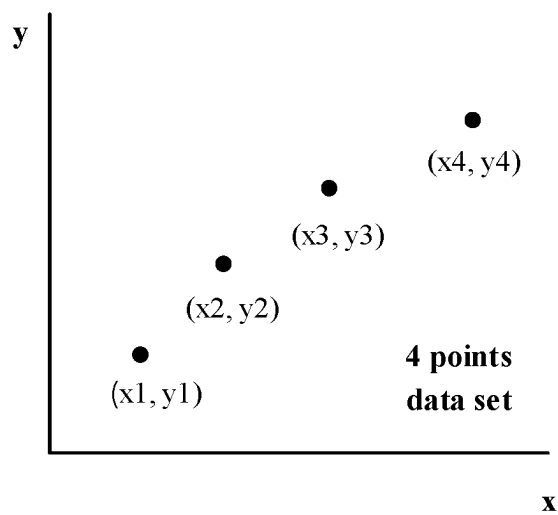


FIG. 7A

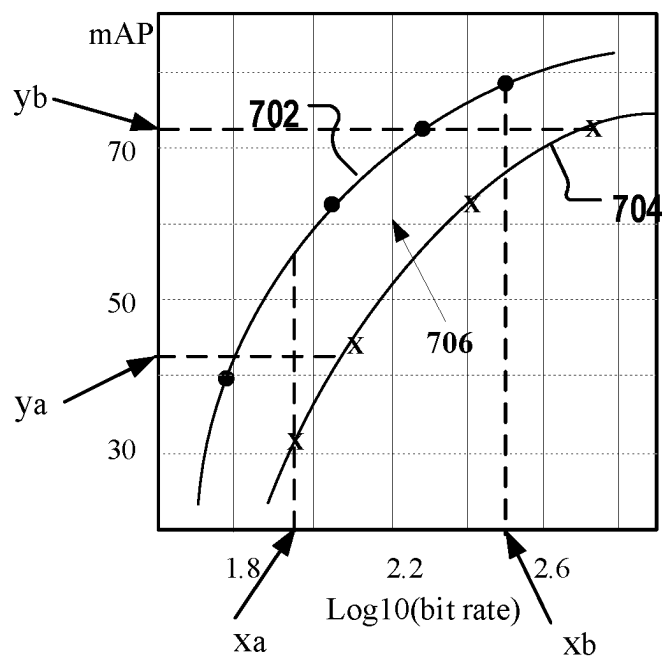


FIG. 7B

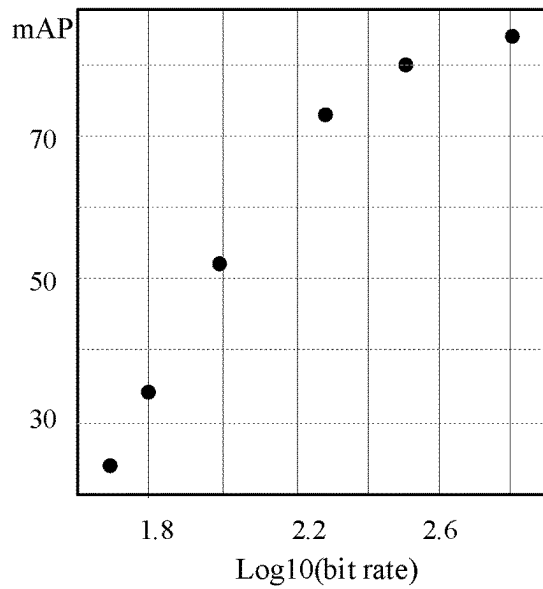


FIG. 8A

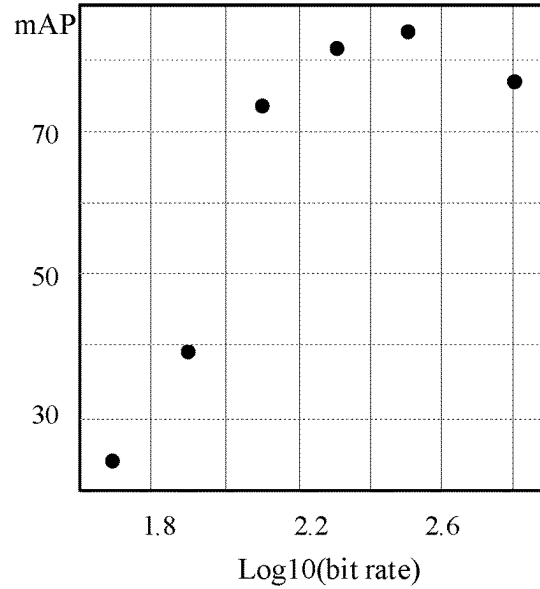


FIG. 8B

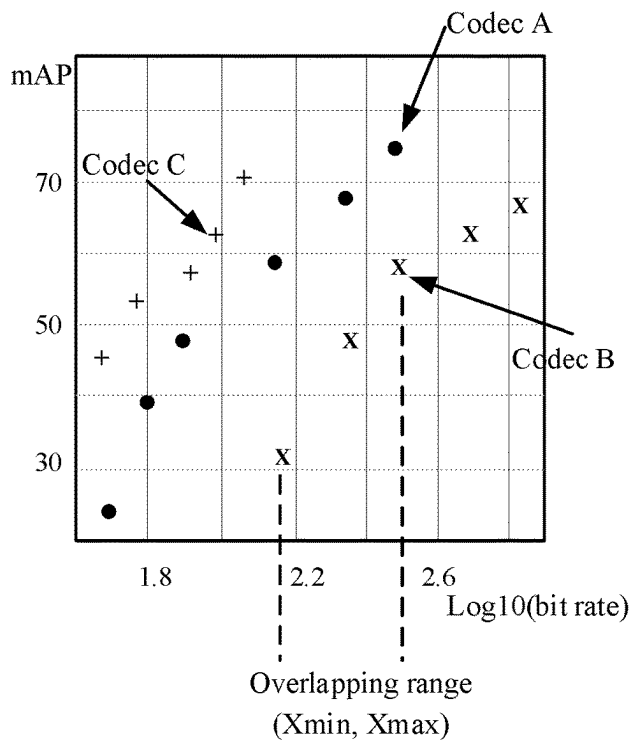


FIG. 8C

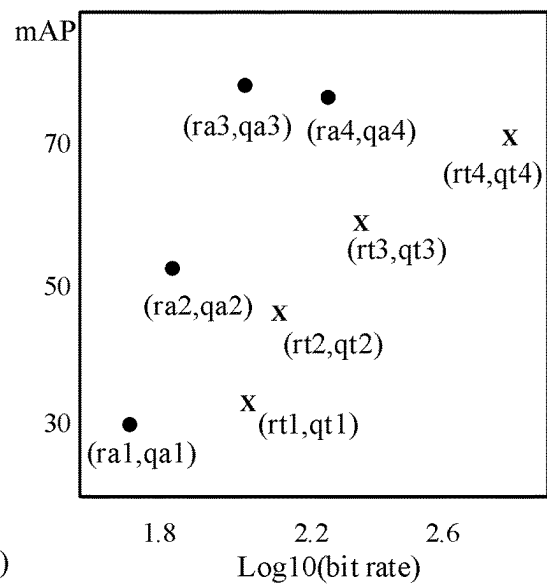
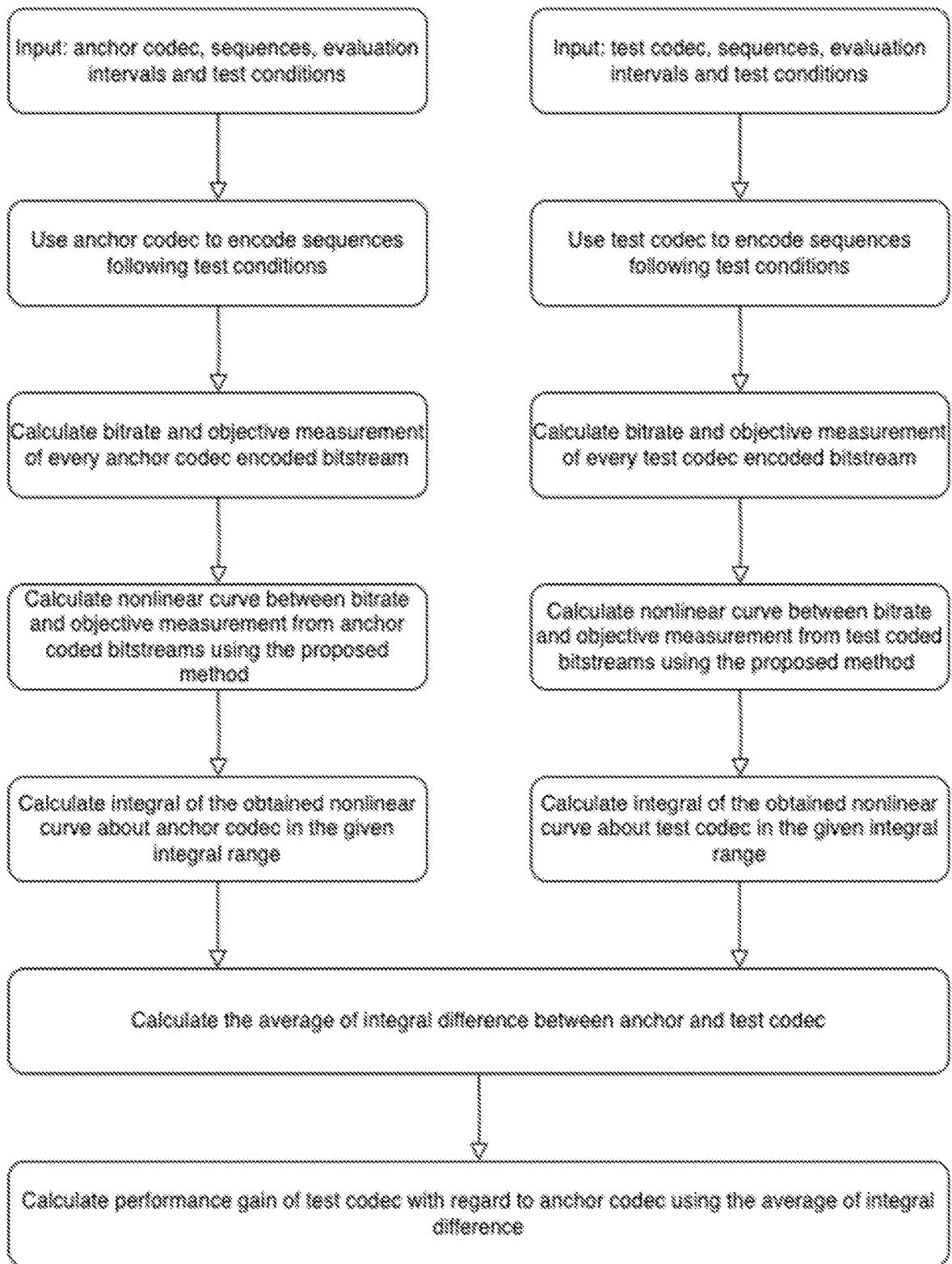


FIG. 8D

**FIG. 9**

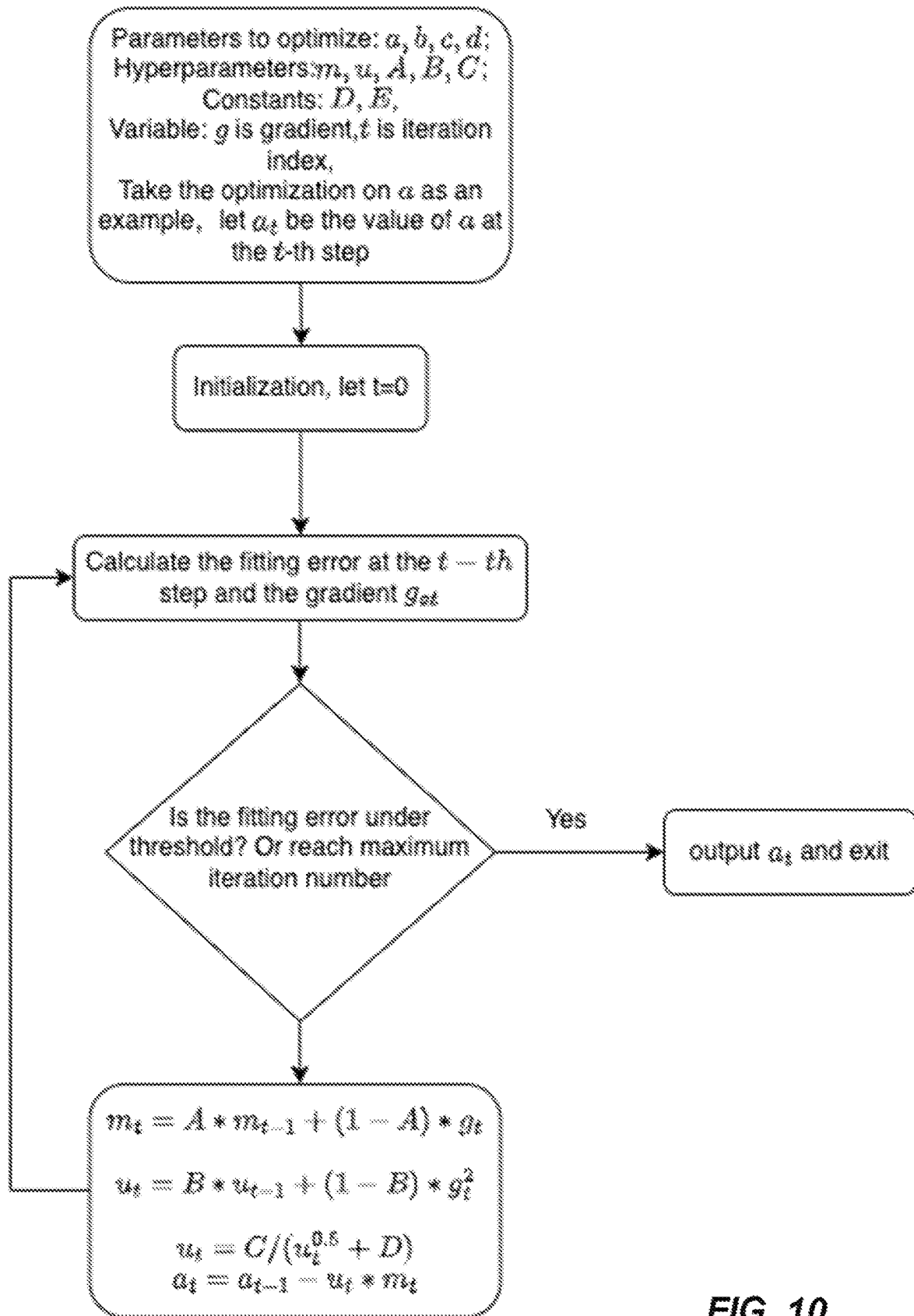


FIG. 10

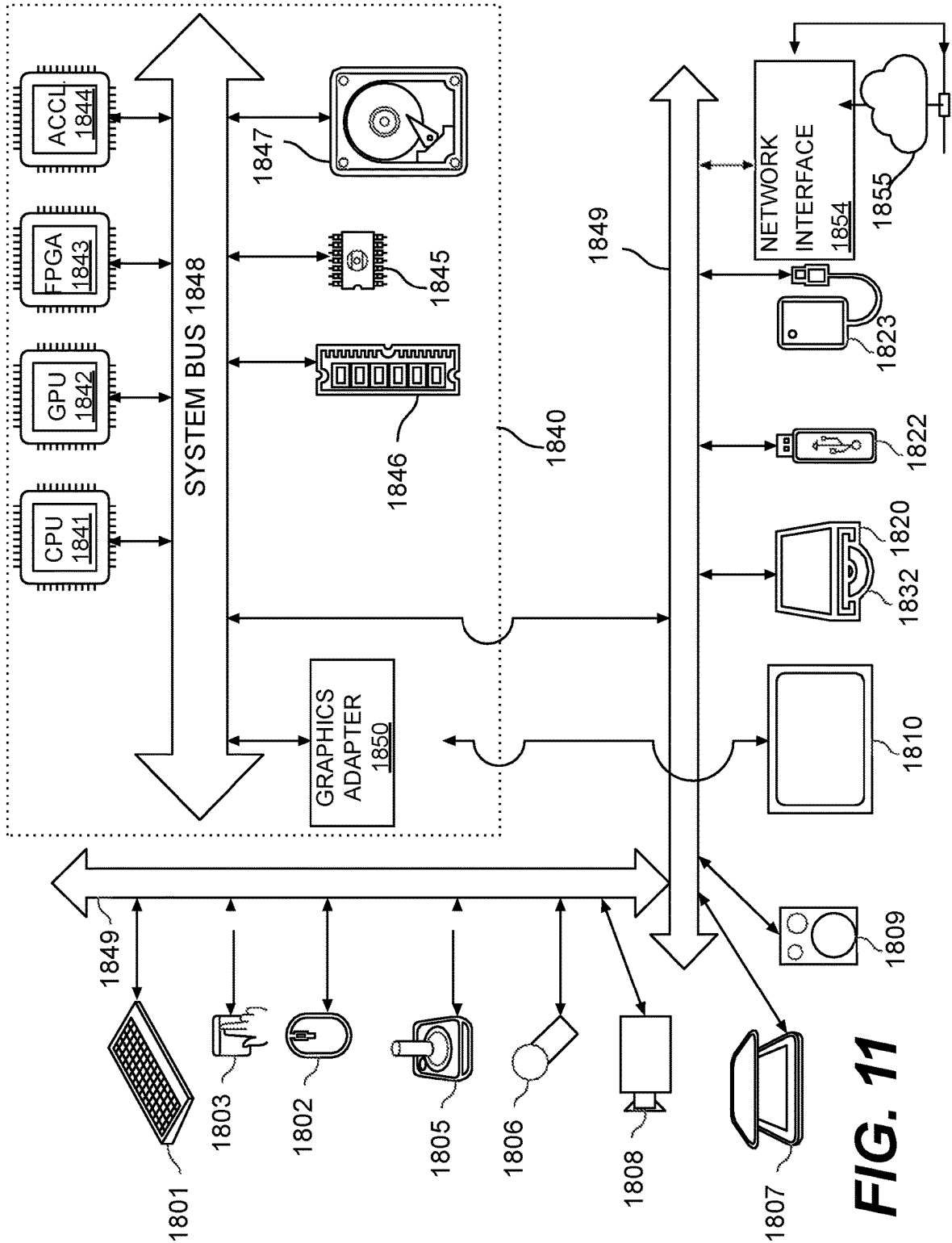


FIG. 11

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METHOD AND APPARATUS FOR VIDEO CODING PERFORMANCE ASSESSMENT

INCORPORATION BY REFERENCE

This application is based on and claims the benefit of priority to U.S. Provisional Application No. 63/527,060, filed on Jul. 16, 2023, which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

This disclosure generally relates to video encoding technologies. More specifically, the disclosed technology involves enhancement on codec performance measurement and evaluation.

BACKGROUND

Over recent decades, video streaming applications have gained significant popularity, driven by the increasing interest for both real-time and on-demand video content across various applications and platforms. Consequently, video streaming now represents the primary source of internet traffic, with an expected further surge in usage due to the proliferation of video-centric applications and advancements in video device capabilities. There is a pressing demand for the development of efficient video compression and delivery algorithms to effectively handle this anticipated growth.

SUMMARY

The present disclosure describes various embodiments of methods, apparatus, and non-transitory computer-readable storage medium for enhancing codec performance measurement and evaluation.

According to one aspect, an embodiment of the present disclosure provides a method for processing video data and evaluating codec performance. The method includes obtaining a first plurality of anchor data points generated based on an anchor video bitstream, wherein: the anchor video bitstream is encoded by an anchor video codec based on a reference video and a corresponding encoding parameter selected from the first plurality of encoding parameters; each anchor data point represents an anchor codec performance using the corresponding encoding parameter, the anchor data point being formatted as a two-tuple including (i) a bit rate or a variation of the bit rate and (ii) a quality measurement; obtaining a second plurality of test data points generated based on a test video bitstream, wherein: the test video bitstream is encoded by a test video codec based on the reference video and a corresponding encoding parameter selected from the second plurality of encoding parameters; each test data point represents a test codec performance using the corresponding encoding parameter, the test data point being formatted as the two-tuple; fitting the first plurality of anchor data points with an anchor curve, the anchor curve being based on an anchor polynomial, wherein the anchor polynomial is monotonic in a first axis range; fitting the second plurality of test data points with a test curve, the test curve being based on a test polynomial, wherein the test polynomial is monotonic in the first axis range; and evaluating the test codec performance based on the anchor curve and the test curve, to obtain an evaluation result.

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According to another aspect, an embodiment of the present disclosure provides an apparatus/device for evaluating video codec. The apparatus/device includes a memory storing instructions; and a processor in communication with the memory. When the processor executes the instructions, the processor is configured to cause the apparatus to perform the above methods for processing video data and evaluating codec performance.

In another aspect, an embodiment of the present disclosure provides non-transitory computer-readable mediums storing instructions which when executed by a computer for video decoding and/or encoding cause the computer to perform the above methods for processing video data and evaluating codec performance.

The above and other aspects and their implementations are described in greater detail in the drawings, the descriptions, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features, the nature, and various advantages of the disclosed subject matter will be more apparent from the following detailed description and the accompanying drawings in which:

FIG. 1 shows a schematic illustration of a simplified block diagram of a computer/communication system (100) in accordance with an example embodiment;

FIG. 2 shows a schematic illustration of a simplified block diagram of a computer/communication system (200) in accordance with an example embodiment;

FIG. 3 shows a schematic illustration of a simplified block diagram of a video decoder in accordance with an example embodiment;

FIG. 4 shows a schematic illustration of a simplified block diagram of a video encoder in accordance with an example embodiment;

FIG. 5 shows a block diagram of a video encoder in accordance with another example embodiment;

FIG. 6 shows a block diagram of a video decoder in accordance with another example embodiment;

FIG. 7A shows example data sets used in Piecewise Cubic Hermit Interpolation (PCHIP);

FIG. 7B illustrates the calculation of average BD-PSNR considering RD curves for two codecs;

FIG. 8A shows data sets with data points following monotonical increasing pattern;

FIG. 8B shows data sets with data points not following monotonical increasing pattern;

FIG. 8C shows overlapping data sets and non-overlapping data sets.

FIG. 8D shows example data sets for two codecs and coordinate for each data point in the data sets.

FIG. 9 shows an example codec evaluation method in accordance with an example embodiment.

FIG. 10 illustrates an exemplary process for solving the polynomial parameters using an iterative parameter optimization procedure

FIG. 11 shows a schematic illustration of a computer system in accordance with example embodiments of this disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

The invention will now be described in detail hereinafter with reference to the accompanied drawings, which form a part of the present invention, and which show, by way of illustration, specific examples of embodiments. Please note

that the invention may, however, be embodied in a variety of different forms and, therefore, the covered or claimed subject matter is intended to be construed as not being limited to any of the embodiments to be set forth below. Please also note that the invention may be embodied as methods, devices, components, or systems. Accordingly, embodiments of the invention may, for example, take the form of hardware, software, firmware or any combination thereof.

Throughout the specification and claims, terms may have nuanced meanings suggested or implied in context beyond an explicitly stated meaning. The phrase “in one embodiment” or “in some embodiments” as used herein does not necessarily refer to the same embodiment and the phrase “in another embodiment” or “in other embodiments” as used herein does not necessarily refer to a different embodiment. Likewise, the phrase “in one implementation” or “in some implementations” as used herein does not necessarily refer to the same implementation and the phrase “in another implementation” or “in other implementations” as used herein does not necessarily refer to a different implementation. It is intended, for example, that claimed subject matter includes combinations of exemplary embodiments/implementations in whole or in part.

In general, terminology may be understood at least in part from usage in context. For example, terms, such as “and”, “or”, or “and/or,” as used herein may include a variety of meanings that may depend at least in part upon the context in which such terms are used. Typically, “or” if used to associate a list, such as A, B or C, is intended to mean A, B, and C, here used in the inclusive sense, as well as A, B or C, here used in the exclusive sense. In addition, the term “one or more” or “at least one” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a”, “an”, or “the”, again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” or “determined by” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

FIG. 1 illustrates a computer network or communication system 100. As shown in FIG. 1, terminal devices may be implemented as servers, personal computers and smart phones but the applicability of the underlying principles of the present disclosure may not be so limited. Embodiments of the present disclosure may be implemented in server or desktop computer 120, laptop computer 110, tablet computer 130 and 140, media players, wearable computers, dedicated video conferencing equipment, and/or the like. The network (150) represents any number or types of networks that convey coded video data among the terminal devices, including for example wireline (wired) and/or wireless communication networks. The communication network (150) may exchange data in circuit-switched, packet-switched, and/or other types of channels. Representative networks include telecommunications networks, local area networks, wide area networks and/or the Internet.

The codec performance measuring and evaluation method provided in this disclosure may be applied to any electronic device, such as server or desktop computer 120, laptop computer 110, tablet computer 130 and 140, to compare the encoding performance of two encoders (also referred to as codec), and further select the codec with better performance

as the actual encoder for use. The corresponding decoder for the encoder can be used in any electronic devices with decoding capabilities, such as TV terminals, PC terminals, mobile terminals, etc.

FIG. 2 illustrates, as an example for an application for the disclosed subject matter, a placement of a video encoder and a video decoder in a video streaming environment. The disclosed subject matter may be equally applicable to other video applications, including, for example, video conferencing, digital TV broadcasting, gaming, virtual reality, storage of compressed video on digital media including CD, DVD, memory stick and the like, and so on. Specifically, the video encoder and/or decoder may be measured, evaluated, and benchmarked with the codec evaluation methods described in this disclosure.

As shown in FIG. 2, a video streaming system may include a video capture subsystem (213) that can include a video source (201), e.g., a digital camera, for creating a stream of video pictures or images (202) that are uncompressed. In an example, the stream of video pictures (202) includes samples that are recorded by a digital camera of the video source (201). The stream of video pictures (202), depicted as a bold line to emphasize a high data volume when compared to encoded video data (204) (or coded video bitstreams), can be processed by an electronic device (220) that includes a video encoder (203) coupled to the video source (201). The video encoder (203) can include hardware, software, or a combination thereof to enable or implement aspects of the disclosed subject matter as described in more detail below. The encoded video data (204) (or encoded video bitstream (204)), depicted as a thin line to emphasize a lower data volume when compared to the stream of uncompressed video pictures (202), can be stored on a streaming server (205) for future use or directly to downstream video devices (not shown). One or more streaming client subsystems, such as client subsystems (206) and (208) in FIG. 2 can access the streaming server (205) to retrieve copies (207) and (209) of the encoded video data (204). A client subsystem (206) can include a video decoder (210), for example, in an electronic device (230). The video decoder (210) decodes the incoming copy (207) of the encoded video data and creates an outgoing stream of video pictures (211) that are uncompressed and that can be rendered on a display (212) (e.g., a display screen) or other rendering devices (not depicted).

FIG. 3 shows a block diagram of a video decoder (310) of an electronic device (330) according to any embodiment of the present disclosure below. The electronic device (330) can include a receiver (331) (e.g., receiving circuitry). The video decoder (310) can be used in place of the video decoder (210) in the example of FIG. 2.

As shown, in FIG. 3, the receiver (331) may receive one or more coded video sequences from a channel (301). To combat network jitter and/or handle playback timing, a buffer memory (315) may be disposed in between the receiver (331) and an entropy decoder/parser (320) (“parser (320)” henceforth). The parser (320) may reconstruct symbols (321) from the coded video sequence. Categories of those symbols include information used to manage operation of the video decoder (310), and potentially information to control a rendering device such as display (312) (e.g., a display screen). The parser (320) may parse/entropy-decode the coded video sequence. The parser (320) may extract from the coded video sequence, a set of subgroup parameters for at least one of the subgroups of pixels in the video decoder. The subgroups can include Groups of Pictures (GOPs), pictures, tiles, slices, macroblocks, Coding Units

(CUs), blocks, Transform Units (TUs), Prediction Units (PUs) and so forth. The parser (320) may also extract from the coded video sequence information such as transform coefficients (e.g., Fourier transform coefficients), quantizer parameter values, motion vectors, and so forth. Reconstruction of the symbols (321) can involve multiple different processing or functional units. The units that are involved and how they are involved may be controlled by the subgroup control information that was parsed from the coded video sequence by the parser (320).

A first unit may include the scaler/inverse transform unit (351). The scaler/inverse transform unit (351) may receive a quantized transform coefficient as well as control information, including information indicating which type of inverse transform to use, block size, quantization factor/parameters, quantization scaling matrices, and the like as symbol(s) (321) from the parser (320). The scaler/inverse transform unit (351) can output blocks comprising sample values that can be input into aggregator (355).

In some cases, the output samples of the scaler/inverse transform (351) can pertain to an intra coded block, i.e., a block that does not use predictive information from previously reconstructed pictures, but can use predictive information from previously reconstructed parts of the current picture. Such predictive information can be provided by an intra picture prediction unit (352). In some cases, the intra picture prediction unit (352) may generate a block of the same size and shape of the block under reconstruction using surrounding block information that is already reconstructed and stored in the current picture buffer (358). The current picture buffer (358) buffers, for example, partly reconstructed current picture and/or fully reconstructed current picture. The aggregator (355), in some implementations, may add, on a per sample basis, the prediction information the intra prediction unit (352) has generated to the output sample information as provided by the scaler/inverse transform unit (351).

In other cases, the output samples of the scaler/inverse transform unit (351) can pertain to an inter coded, and potentially motion compensated block. In such a case, a motion compensation prediction unit (353) can access reference picture memory (357) based on motion vector to fetch samples used for inter-picture prediction. After motion compensating the fetched reference samples in accordance with the symbols (321) pertaining to the block, these samples can be added by the aggregator (355) to the output of the scaler/inverse transform unit (351) (output of unit 351 may be referred to as the residual samples or residual signal) so as to generate output sample information.

The output samples of the aggregator (355) can be subject to various loop filtering techniques in the loop filter unit (356) including several types of loop filters. The output of the loop filter unit (356) can be a sample stream that can be output to the rendering device (312) as well as stored in the reference picture memory (357) for use in future inter-picture prediction.

FIG. 4 shows a block diagram of a video encoder (403) according to an example embodiment of the present disclosure. The video encoder (403) may be included in an electronic device (420). The electronic device (420) may further include a transmitter (440) (e.g., transmitting circuitry). The video encoder (403) can be used in place of the video encoder (403) in the example of FIG. 4.

The video encoder (403) may receive video samples from a video source (401). According to some example embodiments, the video encoder (403) may code and compress the pictures of the source video sequence into a coded video

sequence (443) in real time or under any other time constraints as required by the application. Enforcing appropriate coding speed constitutes one function of a controller (450). In some embodiments, the controller (450) may be functionally coupled to and control other functional units as described below. Parameters set by the controller (450) can include rate control related parameters (picture skip, quantizer, lambda value of rate-distortion optimization techniques, . . .), picture size, group of pictures (GOP) layout, maximum motion vector search range, and the like.

In some example embodiments, the video encoder (403) may be configured to operate in a coding loop. The coding loop can include a source coder (430), and a (local) decoder (433) embedded in the video encoder (403). The decoder (433) reconstructs the symbols to create the sample data in a similar manner as a (remote) decoder would create even though the embedded decoder 433 process coded video stream by the source coder 430 without entropy coding (as any compression between symbols and coded video bitstream in entropy coding may be lossless in the video compression technologies considered in the disclosed subject matter). An observation that can be made at this point is that any decoder technology except the parsing/entropy decoding that may only be present in a decoder also may necessarily need to be present, in substantially identical functional form, in a corresponding encoder. For this reason, the disclosed subject matter may at times focus on decoder operation, which allies to the decoding portion of the encoder. The description of encoder technologies can thus be abbreviated as they are the inverse of the comprehensively described decoder technologies. Only in certain areas or aspects a more detail description of the encoder is provided below.

During operation in some example implementations, the source coder (430) may perform motion compensated predictive coding, which codes an input picture predictively with reference to one or more previously coded picture from the video sequence that were designated as “reference pictures.”

The local video decoder (433) may decode coded video data of pictures that may be designated as reference pictures. The local video decoder (433) replicates decoding processes that may be performed by the video decoder on reference pictures and may cause reconstructed reference pictures to be stored in a reference picture cache (434). In this manner, the video encoder (403) may store copies of reconstructed reference pictures locally that have common content as the reconstructed reference pictures that will be obtained by a far-end (remote) video decoder (absent transmission errors).

The predictor (435) may perform prediction searches for the coding engine (432). That is, for a new picture to be coded, the predictor (435) may search the reference picture memory (434) for sample data (as candidate reference pixel blocks) or certain metadata such as reference picture motion vectors, block shapes, and so on, that may serve as an appropriate prediction reference for the new pictures.

The controller (450) may manage coding operations of the source coder (430), including, for example, setting of parameters and subgroup parameters used for encoding the video data.

Output of all aforementioned functional units may be subjected to entropy coding in the entropy coder (445). The transmitter (440) may buffer the coded video sequence(s) as created by the entropy coder (445) to prepare for transmission via a communication channel (460), which may be a hardware/software link to a storage device which would store the encoded video data. The transmitter (440) may

merge coded video data from the video coder (403) with other data to be transmitted, for example, coded audio data and/or ancillary data streams (sources not shown).

The controller (450) may manage operation of the video encoder (403). During coding, the controller (450) may assign to each coded picture a certain coded picture type, which may affect the coding techniques that may be applied to the respective picture. For example, pictures often may be assigned as one of the following picture types: an Intra Picture (I picture), a predictive picture (P picture), a bi-directionally predictive picture (B Picture), a multiple-predictive picture. Source pictures commonly may be subdivided spatially into a plurality of sample coding blocks as described in further detail below.

FIG. 5 shows a diagram of a video encoder (503) according to another example embodiment of the disclosure. The video encoder (503) is configured to receive a processing block (e.g., a prediction block) of sample values within a current video picture in a sequence of video pictures, and encode the processing block into a coded picture that is part of a coded video sequence. The example video encoder (503) may be used in place of the video encoder (403) in the FIG. 4 example.

For example, the video encoder (503) receives a matrix of sample values for a processing block. The video encoder (503) then determines whether the processing block is best coded using intra mode, inter mode, or bi-prediction mode using, for example, rate-distortion optimization (RDO).

In the example of FIG. 5, the video encoder (503) includes an inter encoder (530), an intra encoder (522), a residue calculator (523), a switch (526), a residue encoder (524), a general controller (521), and an entropy encoder (525) coupled together as shown in the example arrangement in FIG. 5.

The inter encoder (530) is configured to receive the samples of the current block (e.g., a processing block), compare the block to one or more reference blocks in reference pictures (e.g., blocks in previous pictures and later pictures in display order), generate inter prediction information (e.g., description of redundant information according to inter encoding technique, motion vectors, merge mode information), and calculate inter prediction results (e.g., predicted block) based on the inter prediction information using any suitable technique.

The intra encoder (522) is configured to receive the samples of the current block (e.g., a processing block), compare the block to blocks already coded in the same picture, and generate quantized coefficients after transform, and in some cases also to generate intra prediction information (e.g., an intra prediction direction information according to one or more intra encoding techniques).

The general controller (521) may be configured to determine general control data and control other components of the video encoder (503) based on the general control data to, for example, determine the prediction mode of the block and provides a control signal to the switch (526) based on the prediction mode.

The residue calculator (523) may be configured to calculate a difference (residue data) between the received block and prediction results for the block selected from the intra encoder (522) or the inter encoder (530). The residue encoder (524) may be configured to encode the residue data to generate transform coefficients. The transform coefficients are then subject to quantization processing to obtain quantized transform coefficients. In various example embodiments, the video encoder (503) also includes a residual decoder (528). The residual decoder (528) is configured to

perform inverse-transform, and generate the decoded residue data. The entropy encoder (525) may be configured to format the bitstream to include the encoded block and perform entropy coding.

FIG. 6 shows a diagram of an example video decoder (610) according to another embodiment of the disclosure. The video decoder (610) is configured to receive coded pictures that are part of a coded video sequence, and decode the coded pictures to generate reconstructed pictures. In an example, the video decoder (610) may be used in place of the video decoder (410) in the example of FIG. 4.

In the example of FIG. 6, the video decoder (610) includes an entropy decoder (671), an inter decoder (680), a residual decoder (673), a reconstruction module (674), and an intra decoder (672) coupled together as shown in the example arrangement of FIG. 6.

The entropy decoder (671) can be configured to reconstruct, from the coded picture, certain symbols that represent the syntax elements of which the coded picture is made up. The inter decoder (680) may be configured to receive the inter prediction information, and generate inter prediction results based on the inter prediction information. The intra decoder (672) may be configured to receive the intra prediction information, and generate prediction results based on the intra prediction information. The residual decoder (673) may be configured to perform inverse quantization to extract de-quantized transform coefficients, and process the de-quantized transform coefficients to convert the residual from the frequency domain to the spatial domain. The reconstruction module (674) may be configured to combine, in the spatial domain, the residual as output by the residual decoder (673) and the prediction results (as output by the inter or intra prediction modules as the case may be) to form a reconstructed block forming part of the reconstructed picture as part of the reconstructed video.

It is noted that the video encoders (203), (403), and (503), and the video decoders (210), (310), and (610) can be implemented using any suitable technique. In some example embodiments, the video encoders (203), (403), and (503), and the video decoders (210), (310), and (610) can be implemented using one or more integrated circuits. In another embodiment, the video encoders (203), (403), and (503), and the video decoders (210), (310), and (610) can be implemented using one or more processors that execute software instructions.

Online video streaming services, such as YouTube, Netflix, and Vimeo, constitute a significant portion of internet traffic, and their dominance is set to increase further. The surging popularity of video streaming services can be primarily attributed to the widespread availability of higher network bandwidth and improved compression efficiency. This has been facilitated by the proliferation of video playback devices such as smartphones, tablets, and smart TVs. As a result, users now expect seamless streaming capabilities anytime, anywhere, on any device. Video compression emerges as one of the most effective and critical methods to reduce the size of media files, enabling faster and efficient transmission and delivery over networks. Video encoding/decoding technology plays a crucial role in meeting the growing demand for high-quality video content while optimizing bandwidth utilization. Considerable effort is being made towards developing more efficient video codecs.

One task in developing novel video codecs, or improving existing codecs, is measuring, evaluating, and comparing codec performances. Video codec evaluation tool and algorithm evaluate the effectiveness and efficiency of codecs in

encoding and decoding video and/or audio data. The evaluation tool/algorithm is critical in determining a codec's capability to deliver high-quality video output while minimizing the resulting file size and mitigating the demands on data transmission bandwidth. A well-designed and robust evaluation tool is essential to accurately quantify the balance a codec should achieve between preserving video quality and optimizing compression ratios. Gaining a comprehensive, in-depth understanding of codec performance is essential to ensure optimal multimedia experiences across various platforms and applications. Factors such as streaming quality, file storage and bandwidth requirements, and overall user satisfaction are directly influenced by the performance characteristics of the employed codecs. Consequently, thorough codec performance testing plays a pivotal role in delivering seamless and high-quality multimedia content to end-users.

In the research and development process of multimedia codecs, Bjøntegaard Delta Peak Signal-to-Noise Ratio (BD-PSNR) and Bjøntegaard Delta rate (BD-rate) are often used as performance criteria to compare the performance of different codecs. BD-PSNR and BD-BR are used to compare and measure the visual quality difference and bitrate difference, respectively. The higher the BD-PSNR is, the better the encoding quality is. On the other hand, the lower the BD-BR is, the fewer bitrates are required for the encoded bitstream with same or similar quality.

Specifically, BD-PSNR is used to measure the average PSNR gain under the same bit rate conditions, while BD-rate is used to measure the average bit rate savings under the same PSNR conditions. Currently, a commonly used algorithm is based on Piecewise Cubic Hermit Interpolation (PCHIP), which constructs a cubic function using cubic spline interpolation for every two adjacent points. The data between these two points are obtained through interpolation. Here, the process is briefly described with reference to FIG. 7A. In this example, each group of data has four data points as input, denoted as (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , (x_4, y_4) . Taking two points from left to right sequentially, the first step is to interpolate between the first and the second points (x_1, y_1) , (x_2, y_2) . The interpolation polynomial is solved through the input values and the first order derivative values of the two points, then the points between x_1 and x_2 are solved through the interpolation polynomial. The above solution process is repeated sequentially to obtain the piecewise interpolation polynomial in the interval from x_1 to x_4 (i.e., sequentially from x_1 to x_2 , x_2 to x_3 , and x_3 to x_4).

Referring to FIG. 7B for an example implementation for a PCHIP based codec evaluation tool. A benchmark interpolation curve **702** is generated based on data points (or operating points, shown as solid dot points in FIG. 7B) corresponding to a benchmark codec (or referred to as anchor codec), and a test interpolation curve **704** is generated based on data points (marked as "X" in FIG. 7B) corresponding to a test codec. An interpolation curve may also be referred to as a fitting curve, as it is designed to fit the data points. The codec evaluation tool will then calculate the area of the region through the interpolation curve (i.e., the region **706** formed by the two fitting curves, and the two vertical dotted lines), and then calculates BD-PSNR and BD-rate based on the difference in the region between the benchmark fitting curve and the test fitting curve.

Even though commonly used, the existing codec performance evaluation algorithm such as PCHIP, has a few limitations and deficiencies.

First, existing codec evaluation tools require that the input data (data points) follow a monotonically increasing pattern.

When the input data is non-monotonic, the codec evaluation tool fails, resulting in unstable and unreliable evaluation outcomes. Consequently, existing codec evaluation tools mandate that input data strictly adheres to a monotonically increasing pattern, rendering them incapable of handling input data that does not conform to this requirement. This limitation impacts the versatility of codec evaluation tools, as for certain codecs, input data may not consistently follow a monotonic pattern (such as monotonically increasing pattern).

FIG. 8A shows data points that satisfy the monotonicity requirement, whereas FIG. 8B shows the data shown points that do not satisfy the monotonicity requirement. Note that in FIG. 8A and FIG. 8B, the x-axis represents bit rate, such as the bit rate in logarithmic format, and the y-axis represents a quality metric, such as a mean average precision (mAP).

Second, for an existing codec evaluation tool to compare the performance of two codecs, the input data for each codec must have overlap, and the performance evaluation interval is restricted to be the overlapping part of the two set of data points. As shown in FIG. 8C, the solid dot data set represents performance measurements for codec A, while the "X" data set represents performance measurements for codec B. Existing codec evaluation tools may only yield valid evaluation results within the indicated overlapping range (i.e., overlapped bit rate range). Further, for data sets generated from various codecs, the overlapping condition may not always be satisfied. For example, as shown in FIG. 8C, the "+" data set represents performance measurements for codec C, and the "+" data set has no overlap with the "X" data set. In this scenario, existing codec evaluation tools will not be able to evaluate and compare the performance for codec B and codec C.

In this disclosure, various embodiments are described aimed at addressing the aforementioned deficiencies and limitations. These embodiments may support data set that is not monotonically increasing. They also support any given intervals, even there is no overlap, so the codec evaluation tools according to these embodiments offer significant flexibility. These embodiments provide solutions for calculating the BD-PSNR and BD-rate regardless whether the input data follows a monotonical pattern. Therefore, codec evaluation tools according to these embodiments hold considerable practical value and guiding significance.

FIG. 9 shows an example codec evaluation method at a step by step basis.

Input:

The input to the codec evaluation tool may include: a first video codec and a second video codec. Exemplarily, the first video codec may be a benchmark video codec, referred to as an anchor or anchor codec; the second video codec may be the codec that needs to be evaluated, referred to as a test or test codec. The input may further include a set of encoding parameters $[c_1, c_2, \dots, c_n]$, where n is an integer. These parameters may include, for example, spatial and temporal parameters (e.g., resolutions of the video), quantization parameters, coding bitrates, color bit depth, chroma subsampling schemes, etc. The input may further include one or more reference video S , and performance evaluation intervals. The intervals may include a range on the x-axis (the first axis), denoted as $[x_a, x_b]$ (ranging from x_a to x_b , inclusive), or a range on the y-axis (the second axis), denoted as $[y_a, y_b]$. The x-axis may represent a bit rate, or a transformation of bit rate (e.g., bit rate in logarithmic format). The y-axis may represent a quality metric, which may include, but is not limited to, Peak Signal-to-Noise

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Ratio (PSNR), mAP, Multiple Object Tracking Accuracy (MOTA), Bjøntegaard Delta MOTA (BD-MOTA), Structural Similarity Index (SSIM), or Video Quality Metric (VQM). The quality metric may be used to assess the fidelity or perceptual quality of the video content.

In some example implementations, the same set of encoding parameters may be used applied to both the first video codec and the second video codec.

In some example implementations, the first video codec and the second video codec may use different sets of encoding parameters.

In some example implementations, for each encoding parameter, an encoded video bit stream may be encoded by a codec. And a corresponding data point (whether anchor data point or test data point) may be calculated.

In some example implementations, multiple data points may be calculated from a same encoded video bit stream. Output:

The expected output may include the quantitative measure of the performance of the test relative to the anchor. The commonly used indicators may include the average objective quality improvement under the same bit rate (e.g., BD-PSNR), and the average bit rate saving under the same objective quality (e.g., BD-rate).

The following steps may be applicable to any existing codec performance comparison methods or tools.

Step 1:

The first step involves encoding reference video source (denoted as S) separately for each encoding parameter (or namely operation point) using the anchor and the test. This results in two sets of encoded bitstreams to be used as source signal for codec performance evaluation, one set denoted as $[s_{a1}, s_{a2}, \dots, s_{an}]$, and the other as $[s_{t1}, s_{t2}, \dots, s_{tm}]$. Here, “ s_{ak} ” represents the source signal (or encoded bitstream) encoded by the anchor codec from the reference video S using the ck configuration. Similarly, “ s_{tk} ” represents the source signal (or encoded bitstream) encoded by the test codec from the reference video S using the ck configuration. Where k is an integer from 1 to n.

Step 2:

For each of the operation point, the second step is to calculate the bit rate value r and the quality scores q of the encoded source signal. This results in 4 sets of data:

1. $[ra1, ra2, \dots, ran]$: This set represents the bit rate measurements of the source signal encoded by the anchor codec. For example, ra1 is the bit rate measurement of source signal sa1 (encoded by the anchor codec).
2. $[qa1, qa2, \dots, qan]$: This set represents the objective quality measurements of the video encoded by the anchor codec. For example, qa1 is the objective quality measurement of source signal sa1.
3. $[rt1, rt2, \dots, rtm]$: This set represents the bit rate measurements of the source signal encoded by the test codec. For example, rt1 is the bit rate measurement of source signal st1 (encoded by the test codec).
4. $[qt1, qt2, \dots, qtm]$: This set represents the objective quality measurements of the source signal encoded by the test codec. For example, qt1 is the objective quality measurement of source signal st1.

By taking the bit rate measurement r as the x-axis and the objective quality q as the y-axis, these four sets of data collectively form one set of anchor data points for anchor codec performance and one set of test data points for test codec performance. Referring to FIG. 8D for an example, the four solid dots form a set of anchor data points for the anchor codec, while the four “x” form a set of test data

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points for the test codec. Note that these data points are discrete data points. If the two sets of data (represented by “x” and dots) contain points with identical x-axis values, one can determine which codec (anchor or test) performs better by comparing their respective y-axis values. However, in practice, finding such points may be challenging or impossible. Therefore, it is proposed to fit each set of data points (i.e., anchor data point set and test data point set) with a nonlinear relationship fitting curve. This approach enables a quantitative comparison between the anchor and test codecs, facilitating a more meaningful assessment of their performance.

In some example implementations, the set of anchor data points for the anchor codec and the set of test data points for the test codec have overlap (e.g., in x-axis).

In some example implementations, the set of anchor data points for the anchor codec and the set of test data points for the test codec do not have overlap (e.g., in x-axis). In this case, extrapolating at least one data set may be applied.

In some example implementations, when deriving the fitting curves, a specific x-axis value range, such as $[xa, xb]$ may apply. The range may be determined by practical requirements, such as a bitrate range suitable for a particular video streaming environment. It’s important to note that this range may be adjusted to accommodate different use case scenarios.

In some example implementations, when deriving the fitting curves, a specific y-axis value range, such as $[ya, yb]$ may apply. The range may be determined by practical requirements, such as a quality requirement for a particular video streaming environment. It’s important to note that this range may be adjusted to accommodate different use case scenarios.

Step 3:

Numerically fit the anchor data points, that is, data points with $[ra1, ra2, \dots, ran]$ as x-axis values and $[qa1, qa2, \dots, qan]$ as y-axis values (e.g., $(ra1, qa1)$, $(ra2, qa2)$, $\dots$, (ran, qan)). Assume the relationship between r (bitrate, x-axis value) and q (objective quality, y-axis value) can be represented by the following polynomial function:

$$fa(x) = b_0x^3 + b_1x^2 + b_2x + b_3 \quad (1)$$

where b_0, b_1, b_2 , and b_3 are coefficients controlling the shape of the curve, and x is the input parameter (in this example, it corresponds to r), y ($f(x)$) is the output value (in this example, it corresponds to q).

In some example implementations, when numerically fitting the anchor data points, a special constraint for this polynomial may be added which requires its first order derivative to be positive (or non-negative). A further constraint may be added which requires its second order derivative to be negative (or non-positive). These constraints may be imposed in a given x range (e.g., a bit rate range). The first constraint may ensure the fitting curve to be monotonic increasing, and the second constraint may ensure the fitting curve to be convex. Another special constraint for this polynomial is that its first order derivative to be positive in the given x range, which ensures the curve to be monotonic increasing. Another special constraint for this polynomial is that its first order derivative to be negative in the given x range, which ensures the curve to be monotonic decreasing. The first order derivative is:

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$$f'(x) = 3b_0x^2 + 2b_1x + b_2 \quad (2)$$

and the second derivative is:

$$f''(x) = 6b_0x + 2b_1 \quad (3)$$

In some example implementations, the first order derivative of the polynomial needs to be negative.

In some example implementations, a further constraint is imposed, such that $f(\min(x)) > 0$ and $f(\max(x)) < 100$, where $\min(x)$ and $\max(x)$ represent the minimum and maximum values of x , respectively.

The fitting curve could be obtained by solving, for example, a typical nonlinear curve-fitting problem. For example, parameters b_0, b_1, b_2, b_3 may be obtained by minimizing least square error between observed values and fitted values with any suitable optimization method.

As an example, let (x_i, y_i) for i in $[1, 2, \dots, N]$ denote the bitrate and metric value of a data point (e.g., a data point as shown in FIG. 8D). $N=6$ is the number of encoding parameters (e.g., 6 Quantization Parameters (QP)). The least square error between observed values and fitted values with any suitable optimization method may be solved using equation below:

$$\min_{b_0, b_1, b_2, b_3} \sum_{i \in [1, 2, \dots, N]} [y_i - f(x_i)]^2 \quad (4)$$

As another example, FIG. 10 illustrates an exemplary process for solving the parameters b_0, b_1, b_2, b_3 using an iterative parameter optimization procedure. In FIG. 10, the to be optimized parameters a, b, c , and d correspond to parameters b_0, b_1, b_2 , and b_3 in the above polynomial function, respectively.

After applying the optimization process described above, we obtain the nonlinear curve, i.e. $fa(x)$, between the anchor bitrates and quality measurements, i.e., $[ra_1, ra_2, \dots, ra_m]$ and $[qa_1, qa_2, \dots, qa_m]$. FIG. 7B shows an example nonlinear curve 704 as the fitting curve for the anchor codec data points (denoted by "X").

Step 4:

For the test data, i.e., $[rt_1, rt_2, \dots, rt_m]$ and $[qt_1, qt_2, \dots, qt_m]$, repeat the nonlinear relationship curve solving process as describe in step 3 for the anchor codec. We obtain the nonlinear relationship curve, $ft(x)$, between $[rt_1, rt_2, \dots, rt_m]$ and $[qt_1, qt_2, \dots, qt_m]$. FIG. 7B shows an example nonlinear curve 702 as the fitting curve for the test codec data points (denoted by solid dot points).

The fitting curve derived with the aforementioned constraints can better capture the nonlinear relationship between bitrate and performance indicators (quality metrics). Specifically, it has the following characteristics: 1) At the middle bitrate points, there is an almost linear relationship between the performance indicators and the bitrate, and the performance indicators rise rapidly with the increase in bitrate. 2) When the bitrate increases to a certain extent, further increasing the bitrate may not bring additional gains in performance indicators, that is, the performance indicators enter a saturation area. 3) When the bitrate is reduced to a certain level, the performance indicators will not decrease with the reduction in bitrate.

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Further, the constrained cubic curve-fitting method as proposed has the following advantages: 1) It gives valid values for all test cases as long as there are overlapping regions (e.g., overlapping x -axis values) between the two curves. 2) It is more transparent and consistent as all data points may be used for evaluation, compared to other methods, such as the BDEExtend (pareto) method. 3) The cubic curve-fitting method achieves smaller fitting error than other methods, such as the BDEExtend (pareto) method, therefore, it may provide more accurate quality metrics values, such as BD-rate values.

Step 5:

For the given performance evaluation interval $[xa, xb]$, calculate the difference between integral area of $fa(x)$ (i.e., curve 704 in FIG. 7B) and $ft(x)$ (i.e., curve 702 in FIG. 7B) within the evaluation interval. This result will be denoted as BD-PSNR, which represents the average objective PSNR gain under the same bitrate. BD-PSNR may be calculated using equation below:

$$BD-PSNR = \frac{1}{xb - xa} \int_{xa}^{xb} [ft(x) - fa(x)] dx \quad (5)$$

Refer to FIG. 7B for an example interval $[xa, xb]$.

Step 6:

For the given performance evaluation interval $[ya, yb]$, calculate the difference between the integral area of $fa(x)$ (i.e., curve 704 in FIG. 7B) and $ft(x)$ (i.e., curve 702 in FIG. 7B) within the evaluation interval $[ya, yb]$. This result can further be converted into BD-rate, which represents the average bitrate gain under the same objective PSNR. BD-rate may be calculated using equation below:

$$BD-rate = 10 \frac{1}{yb - ya} \int_{ya}^{yb} [ft(y) - fa(y)] dy - 1 \quad (6)$$

Refer to FIG. 7B for an example interval $[ya, yb]$.

In some embodiments, the codec performance evaluation method described above may be applied in video streaming applications. In such applications, a video server may serve (e.g., transmit video stream to) multiple end users simultaneously. Depending on the transmitting channel/link, the bit rate for the video streaming may be different. For example, user A may be covered by excellent WiFi signal and the link speed is over 200 Mbps (megabits per second), user B is under cellular coverage and the link speed may be around 50 Mbps, user C is under poor coverage and the link speed is only around 10 Mbps. For a same video source, there may be multiple versions of coded bitstreams, and these bitstreams may be encoded by different codecs. The codec performance evaluation tool according to this disclosure may be used to compare 3 codecs, with 3 bit rate ranges (in Mbps): $[0, 20]$, $[20, 100]$, and $[100, 300]$. After the comparison, it turns out that:

Codec A outperforms in the bit rate range $[0, 20]$;

Codec B outperforms in the bit rate range $[20, 100]$; and

Codec C outperforms in the bit rate range $[100, 300]$.

Therefore, based on each user's coverage condition, the video stream server may serve:

User A (link speed 200 Mbps) with bitstream encoded by codec C;

User B (link speed 50 Mbps) with bitstream encoded by codec B; and

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User C (link speed 10 Mbps) with bitstream encoded by codec A.

In some embodiments, the similar concept may apply to an individual user with varying network coverage condition caused by, for example, user movement. When the user is moving, the available bandwidth may change due to factors like changing cellular signal strength or network coverage conditions. In such scenarios, a video streaming platform dynamically switch the video bitstream based on the codec used for the encoding. Specifically, a video streaming platform may employ a dynamic approach to deliver an optimal viewing experience based on the user's available bandwidth/link speed. By leveraging multiple encoded bitstreams (for same video source, for example, a same movie), the platform may intelligently select a bitstream encoded by a codec that outperforms other codecs within the specific bit rate range corresponding to the user's current link speed. This adaptive strategy ensures that the chosen bitstream is compressed using the most efficient codec for the given bandwidth constraints, thereby delivering the maximizing video quality which is adapted to the current link speed. Stated another way, for a same video chunk, multiple video bitstreams may be encoded by different codecs. Using the coded evaluation tool/algorithm provided in this disclosure, it is possible to select an optimal bitstream encoded by the codec that outperforms other codecs in a specific bit rate range, and the specific bit rate range is determined by the bandwidth of a video streaming session.

An exemplary method following the principles underlying the implementations above for measuring and evaluating codec performance may include a portion or all of the following steps: step 1: obtaining m anchor data points each generated based on a respective anchor encoded video bitstream, m being an integer, wherein: the respective anchor encoded video bitstream is a bitstream encoded by an anchor video codec based on a reference video and a corresponding encoding parameter selected from m encoding parameters; each of the m anchor data points represents an anchor codec performance using the corresponding encoding parameter and is a two-tuple formed by an x -axis value representing a bit rate or a variation of the bit rate and a y -axis value representing a quality measurement; step 2: obtaining n test data points each generated based on a respective encoded test video bitstream, n being an integer, wherein: the respective test encoded video bitstream is a bitstream encoded by a test video codec based on the reference video and a corresponding encoding parameter selected from n encoding parameters; each of the n test data points represents a test codec performance using the corresponding encoding parameter and is a two-tuple formed by an x -axis value representing a bit rate or a variation of the bit rate and a y -axis value representing a quality measurement; step 3: fitting the m anchor data points with an anchor curve, the anchor curve being based on an anchor polynomial, wherein the anchor polynomial is monotonic in an x -axis range; step 4: fitting the n test data points with a test curve, the anchor curve being based on a test polynomial, wherein the test polynomial is monotonic in the x -axis range; and step 5: evaluating the test codec performance based on the anchor curve and the test curve, to obtain an evaluation result.

In any portion or combination of the implementations above, at least one of following conditions satisfies: the m anchor data points are monotonically increasing; the m anchor data points are not monotonically increasing; the n test data points are monotonically increasing; or the n test data points are not monotonically increasing.

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In any portion or combination of the implementations above, the m encoding parameters are the same as the n encoding parameters and m equals to n .

In any portion or combination of the implementations above, at least one of following conditions satisfies: an x -axis range of the m anchor data points and an x -axis range of the n test data points have overlap; or an x -axis range of the m anchor data points and an x -axis range of the n test data points have no overlap.

In any portion or combination of the implementations above, each of the anchor polynomial and the test polynomial is monotonic increasing.

The embodiments in this disclosure may be used to evaluation and compare two codecs, for example, a first codec and a second codec. In some example implementations, the first codec may be an anchor codec (benchmark codec), and the second codec may be a test codec. Specifically, the test codec may be a codec under development, and/or in a standardization process to be adopted to a video encoding/decoding standard.

Operations above may be combined or arranged in any amount or order, as desired. Two or more of the steps and/or operations may be performed in parallel. Embodiments and implementations in the disclosure may be used separately or combined in any order. Steps in one embodiment/method may be split to form multiple sub-methods, each of the sub-methods may be independent of other steps in the embodiment and may form a standalone solution. Further, each of the methods (or embodiments) may be performed by a device, and the device may be implemented by processing circuitry (e.g., one or more processors or one or more integrated circuits). In one example, the one or more processors execute a program that is stored in a non-transitory computer-readable medium.

The techniques described above, can be implemented as computer software using computer-readable instructions and physically stored in one or more computer-readable media. For example, FIG. 13 shows a computer system (1800) suitable for implementing certain embodiments of the disclosed subject matter.

The computer software can be coded using any suitable machine code or computer language, that may be subject to assembly, compilation, linking, or like mechanisms to create code comprising instructions that can be executed directly, or through interpretation, micro-code execution, and the like, by one or more computer central processing units (CPUs), Graphics Processing Units (GPUs), and the like.

The instructions can be executed on various types of computers or components thereof, including, for example, personal computers, tablet computers, servers, smartphones, gaming devices, internet of things devices, and the like.

The components shown in FIG. 13 for computer system (1800) are exemplary in nature and are not intended to suggest any limitation as to the scope of use or functionality of the computer software implementing embodiments of the present disclosure. Neither should the configuration of components be interpreted as having any dependency or requirement relating to any one or combination of components illustrated in the exemplary embodiment of a computer system (1800).

Computer system (1800) may include certain human interface input devices. Input human interface devices may include one or more of (only one of each depicted): keyboard (1801), mouse (1802), trackpad (1803), touch screen (1810), data-glove (not shown), joystick (1805), microphone (1806), scanner (1807), camera (1808).

Computer system (1800) may also include certain human interface output devices. Such human interface output devices may be stimulating the senses of one or more human users through, for example, tactile output, sound, light, and smell/taste. Such human interface output devices may include tactile output devices (for example tactile feedback by the touch-screen (1810), data-glove (not shown), or joystick (1805), but there can also be tactile feedback devices that do not serve as input devices), audio output devices (such as: speakers (1809), headphones (not depicted)), visual output devices (such as screens (1810) to include CRT screens, LCD screens, plasma screens, OLED screens, each with or without touch-screen input capability, each with or without tactile feedback capability—some of which may be capable to output two dimensional visual output or more than three dimensional output through means such as stereographic output; virtual-reality glasses (not depicted), holographic displays and smoke tanks (not depicted)), and printers (not depicted).

Computer system (1800) can also include human accessible storage devices and their associated media such as optical media including CD/DVD ROM/RW (1820) with CD/DVD or the like media (1821), thumb-drive (1822), removable hard drive or solid state drive (1823), legacy magnetic media such as tape and floppy disc (not depicted), specialized ROM/ASIC/PLD based devices such as security dongles (not depicted), and the like.

Those skilled in the art should also understand that term “computer readable media” as used in connection with the presently disclosed subject matter does not encompass transmission media, carrier waves, or other transitory signals.

Computer system (1800) can also include an interface (1854) to one or more communication networks (1855). Networks can for example be wireless, wireline, optical. Networks can further be local, wide-area, metropolitan, vehicular and industrial, real-time, delay-tolerant, and so on. Examples of networks include local area networks such as Ethernet, wireless LANs, cellular networks to include GSM, 3G, 4G, 5G, LTE and the like, TV wireline or wireless wide area digital networks to include cable TV, satellite TV, and terrestrial broadcast TV, vehicular and industrial to include CAN bus, and so forth.

Aforementioned human interface devices, human-accessible storage devices, and network interfaces can be attached to a core (1840) of the computer system (1800).

The core (1840) can include one or more Central Processing Units (CPU) (1841), Graphics Processing Units (GPU) (1842), specialized programmable processing units in the form of Field Programmable Gate Areas (FPGA) (1843), hardware accelerators for certain tasks (1844), graphics adapters (1850), and so forth. These devices, along with Read-only memory (ROM) (1845), Random-access memory (1846), internal mass storage such as internal non-user accessible hard drives, SSDs, and the like (1847), may be connected through a system bus (1848). In some computer systems, the system bus (1848) can be accessible in the form of one or more physical plugs to enable extensions by additional CPUs, GPU, and the like. The peripheral devices can be attached either directly to the core's system bus (1848), or through a peripheral bus (1849). In an example, the screen (1810) can be connected to the graphics adapter (1850). Architectures for a peripheral bus include PCI, USB, and the like.

The computer readable media can have computer code thereon for performing various computer-implemented operations. The media and computer code can be those specially designed and constructed for the purposes of the

present disclosure, or they can be of the kind well known and available to those having skill in the computer software arts.

While this disclosure has described several exemplary embodiments, there are alterations, permutations, and various substitute equivalents, which fall within the scope of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise numerous systems and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope thereof.

What is claimed is:

1. A method for processing video data, comprising:

obtaining a first plurality of anchor data points generated based on an anchor video bitstream, wherein:

the anchor video bitstream is encoded by an anchor video codec based on a reference video and a corresponding encoding parameter selected from the first plurality of encoding parameters;

each anchor data point represents an anchor codec performance using the corresponding encoding parameter, the anchor data point being formatted as a two-tuple including (i) a bit rate or a variation of the bit rate and (ii) a quality measurement;

obtaining a second plurality of test data points generated based on a test video bitstream, wherein:

the test video bitstream is encoded by a test video codec based on the reference video and a corresponding encoding parameter selected from the second plurality of encoding parameters;

each test data point represents a test codec performance using the corresponding encoding parameter, the test data point being formatted as the two-tuple;

fitting the first plurality of anchor data points with an anchor curve, the anchor curve being based on an anchor polynomial, wherein the anchor polynomial is monotonic in a first axis range;

fitting the second plurality of test data points with a test curve, the test curve being based on a test polynomial, wherein the test polynomial is monotonic in the first axis range; and

evaluating the test codec performance based on the anchor curve and the test curve, to obtain an evaluation result.

2. The method of claim 1, wherein at least one of following conditions satisfies:

the first plurality of anchor data points are monotonically increasing;

the first plurality of anchor data points are not monotonically increasing;

the second plurality of test data points are monotonically increasing; or

the second plurality of test data points are not monotonically increasing.

3. The method of claim 1, wherein the first plurality of encoding parameters are the same as the second plurality of encoding parameters.

4. The method of claim 1, wherein at least one of following conditions satisfies:

an first axis range of the first plurality of anchor data points and an first axis range of the second plurality of test data points have overlap; or

an first axis range of the first plurality of anchor data points and an first axis range of the second plurality of test data points have no overlap.

5. The method of claim 1, wherein each of the anchor polynomial and the test polynomial is monotonic increasing.

6. The method of claim 1, wherein:

the anchor polynomial is represented by a following equation:

$$fa(x) = b_0x^3 + b_1x^2 + b_2x + b_3$$

where: b0, b1, b2, and b3 are coefficients, an x-axis of the equation represents a bit rate or a variation of the bit rate, and a y-axis of the equation represents a quality measurement; and

the test polynomial is represented by a following equation:

$$ft(x) = c_0x^3 + c_1x^2 + c_2x + c_3$$

where: c0, c1, c2, and c3 are coefficients, an x-axis of the equation represents a bit rate or a variation of the bit rate, and a y-axis of the equation represents a quality measurement.

7. The method of claim 6, further comprising:

deriving the anchor polynomial with a constraint such that within the first axis range, a first order derivative of the anchor polynomial is positive, wherein the first order derivative of the anchor polynomial is represented by a following equation:

$$f'(x) = 3b_0x^2 + 2b_1x + b_2;$$

and

deriving the anchor polynomial with a constraint such that within the first axis range, a first order derivative of the test polynomial is positive, wherein the first order derivative of the test polynomial is represented by a following equation:

$$f'(x) = 3c_0x^2 + 2c_1x + c_2.$$

8. The method of claim 7, further comprising:

deriving the anchor polynomial with a further constraint such that within the first axis range, a second order derivative of the anchor polynomial is negative; and

deriving the test polynomial with a further constraint such that a second order derivative of the test polynomial is negative.

9. The method of claim 7, further comprising deriving a Bjøntegaard Delta Peak Signal-to-Noise Ratio (BD-PSNR) using an equation below:

$$BD-PSNR = \frac{1}{xb - xa} \int_{xa}^{xb} [ft(x) - fa(x)]dx$$

where xa and xb are a start and an end of the first axis range, respectively.

10. The method of claim 7, further comprising deriving a Bjøntegaard Delta Rate (BD-rate) using an equation below:

$$BD-rate = 10 \frac{1}{yb - ya} \int_{ya}^{yb} [ft(y) - fa(y)]dy - 1$$

where ya and yb are a start and an end of a quality measurement interval in the y-axis.

11. The method of claim 1, further comprising:

in response to the evaluation result indicating that the anchor video codec outperforms the test video codec within the first axis range and a link speed of a video streaming session falling into the first axis range, selecting an encoded video bitstream encoded by the anchor video codec for the video streaming session; and

in response to the evaluation result indicating that the test video codec outperforms the anchor video codec within the first axis range and a link speed of a video streaming session falling into first axis range, selecting an encoded video bitstream encoded by the test video codec for the video streaming session.

12. The method of claim 11, further comprising:

transmitting the selected encoded video bitstream in the video streaming session.

13. A device for processing video data, the device comprising a memory for storing computer instructions and a processor in communication with the memory, wherein, when the processor executes the computer instructions, the processor is configured to cause the device to:

obtain a first plurality of anchor data points generated based on an anchor video bitstream, wherein:

the anchor video bitstream is encoded by an anchor video codec based on a reference video and a corresponding encoding parameter selected from the first plurality of encoding parameters;

each anchor data point represents an anchor codec performance using the corresponding encoding parameter, the anchor data point being formatted as a two-tuple including (i) a bit rate or a variation of the bit rate and (ii) a quality measurement;

obtain a second plurality of test data points generated based on a test video bitstream, wherein:

the test video bitstream is encoded by a test video codec based on the reference video and a corresponding encoding parameter selected from the second plurality of encoding parameters;

each test data point represents a test codec performance using the corresponding encoding parameter, the test data point being formatted as the two-tuple;

fit the first plurality of anchor data points with an anchor curve, the anchor curve being based on an anchor polynomial, wherein the anchor polynomial is monotonic in a first axis range;

fit the second plurality of test data points with a test curve, the test curve being based on a test polynomial, wherein the test polynomial is monotonic in the first axis range; and

evaluate the test codec performance based on the anchor curve and the test curve, to obtain an evaluation result.

14. The device of claim 13, wherein at least one of following conditions satisfies:

the first plurality of anchor data points are monotonically increasing;

the first plurality of anchor data points are not monotonically increasing;

the second plurality of test data points are monotonically increasing; or

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the second plurality of test data points are not monotonically increasing.

15. The device of claim 13, wherein at least one of following conditions satisfies:

an first axis range of the first plurality of anchor data points and an first axis range of the second plurality of test data points have overlap; or

an first axis range of the first plurality of anchor data points and an first axis range of the second plurality of test data points have no overlap.

16. The device of claim 13, wherein each of the anchor polynomial and the test polynomial is monotonic increasing.

17. The device of claim 13, wherein:

the anchor polynomial is represented by a following equation:

$$fa(x) = b_0x^3 + b_1x^2 + b_2x + b_3$$

where: b0, b1, b2, and b3 are coefficients, an x-axis of the equation represents a bit rate or a variation of the bit rate, and a y-axis of the equation represents a quality measurement; and

the test polynomial is represented by a following equation:

$$ft(x) = c_0x^3 + c_1x^2 + c_2x + c_3$$

where: c0, c1, c2, and c3 are coefficients, an x-axis of the equation represents a bit rate or a variation of the bit rate, and a y-axis of the equation represents a quality measurement.

18. The device of claim 17, wherein, when the processor executes the computer instructions, the processor is configured to further cause the device to:

derive the anchor polynomial with a constraint such that within the first axis range, a first order derivative of the anchor polynomial is positive, wherein the first order derivative of the anchor polynomial is represented by a following equation:

$$f'(x) = 3b_0x^2 + 2b_1x + b_2;$$

and

derive the anchor polynomial with a constraint such that within the first axis range, a first order derivative of the

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test polynomial is positive, wherein the first order derivative of the test polynomial is represented by a following equation:

$$f'(x) = 3c_0x^2 + 2c_1x + c_2.$$

19. The device of claim 18, when the processor executes the computer instructions, the processor is configured to further cause the device to:

derive the anchor polynomial with a further constraint such that within the first axis range, a second order derivative of the anchor polynomial is negative; and

derive the test polynomial with a further constraint such that a second order derivative of the test polynomial is negative.

20. A non-transitory storage medium for storing computer readable instructions, the computer readable instructions, when executed by a processor, causing the processor to:

obtain a first plurality of anchor data points generated based on an anchor video bitstream, wherein:

the anchor video bitstream is encoded by an anchor video codec based on a reference video and a corresponding encoding parameter selected from the first plurality of encoding parameters;

each anchor data point represents an anchor codec performance using the corresponding encoding parameter, the anchor data point being formatted as a two-tuple including (i) a bit rate or a variation of the bit rate and (ii) a quality measurement;

obtain a second plurality of test data points generated based on a test video bitstream, wherein:

the test video bitstream is encoded by a test video codec based on the reference video and a corresponding encoding parameter selected from the second plurality of encoding parameters;

each test data point represents a test codec performance using the corresponding encoding parameter, the test data point being formatted as the two-tuple;

fit the first plurality of anchor data points with an anchor curve, the anchor curve being based on an anchor polynomial, wherein the anchor polynomial is monotonic in a first axis range;

fit the second plurality of test data points with a test curve, the test curve being based on a test polynomial, wherein the test polynomial is monotonic in the first axis range; and

evaluate the test codec performance based on the anchor curve and the test curve, to obtain an evaluation result.

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